📚 Project Reference List

Hybrid Chemical-ML Modelling of Fluoride Adsorption Using Coconut Husk Activated Carbon

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Category 1 — Foundation: Langmuir Isotherm Theory

[R1] The Original Langmuir Paper (1918) — PUBLIC DOMAIN ✅ FREE

> Langmuir, I. (1918). **The adsorption of gases on plane surfaces of glass, mica and platinum.** *Journal of the American Chemical Society*, 40(9), 1361–1403.

- **DOI:** 10.1021/ja02242a004
- **Direct link:** https://doi.org/10.1021/ja02242a004
- **Free PDF (Semantic Scholar):** https://www.semanticscholar.org/paper/THE-ADSORPTION-OF-GASES-ON-PLANE-SURFACES-OF-GLASS,-Langmuir/7e20841680a7f70b0b8d903bf15ac769d61c519c
- **Status:** ✅ PUBLIC DOMAIN (1918 — copyright expired)
- **Used for:** $q_e = q_{max} \times K_L \times C_e / (1 + K_L \times C_e)$ formula, monolayer adsorption theory

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[R2] Freundlich Isotherm Comparison

> Freundlich, H. (1906). **Über die Adsorption in Lösungen.** *Zeitschrift für Physikalische Chemie*, 57, 385–470.

- **Status:** ✅ PUBLIC DOMAIN (1906)
- **Used for:** Comparison with Langmuir model; Langmuir selected as superior fit for monolayer systems

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[R3] Langmuir Isotherm Validation — Modern Review

> Foo, K.Y., & Hameed, B.H. (2010). **Insights into the modeling of adsorption isotherm systems.** *Chemical Engineering Journal*, 156(1), 2–10.

- **DOI:** 10.1016/j.cej.2009.09.013
- **Journal link:** https://www.sciencedirect.com/science/article/abs/pii/S1385894709007669
- **ResearchGate:** https://www.researchgate.net/publication/222444248
- **Access:** Elsevier (paywalled); check institution or ResearchGate
- **Used for:** Validation of Langmuir model selection criteria and parameter interpretation

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Category 2 — Adsorption Kinetics

[R4] Pseudo-Second-Order Kinetics — Most Cited Kinetics Paper

> Ho, Y.S., & McKay, G. (1999). **Pseudo-second order model for sorption processes.** *Process Biochemistry*, 34(5), 451–465.

- **DOI:** 10.1016/S0032-9592(98)00112-5
- **Journal link:** https://www.sciencedirect.com/science/article/abs/pii/S0032959298001125
- **HKUST repository:** https://researchportal.hkust.edu.hk/en/publications/pseudo-second-order-model-for-sorption-processes/
- **Access:** Elsevier (paywalled); often available via ResearchGate
- **Used for:** PSO kinetic model: $dq/dt = k_2(qe - q)^2$; kinetic_factor calculation in simulation

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[R5] Pseudo-First-Order Kinetics — Lagergren Model

> Lagergren, S. (1898). **About the theory of so-called adsorption of soluble substances.** *Kungliga Svenska Vetenskapsakademiens Handlingar*, 24, 1–39.

- **Status:** ✅ PUBLIC DOMAIN (1898)
- **Used for:** Pseudo-first-order comparison; Ce = C0 × exp(−k × Dose × Time) term in simulation

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[R6] Kinetic Modelling Critical Review

> Simonin, J.P. (2016). **On the comparison of pseudo-first order and pseudo-second order rate laws in the modeling of adsorption kinetics.** *Chemical Engineering Journal*, 300, 254–263.

- **DOI:** 10.1016/j.cej.2016.04.079
- **Journal link:** https://www.sciencedirect.com/science/article/abs/pii/S1385894716305319
- **Access:** Elsevier (paywalled)
- **Used for:** Justification of PSO over PFO for fluoride systems

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Category 3 — Fluoride Adsorption on Coconut Husk / Activated Carbon

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[R7] Key Study — Coconut Husk Activated Carbon Fixed Bed Column ✅ RESEARCHGATE FREE

> Talat, M., Mohan, S., Dixit, V., Singh, D.K., Hasan, S.H., & Srivastava, O.N. (2018). **Effective removal of fluoride from water by coconut husk activated carbon in fixed bed column: Experimental and breakthrough curves analysis.** *Groundwater for Sustainable Development*, 7, 48–55.

- **DOI:** 10.1016/j.gsd.2018.03.001
- **Journal link:** https://www.sciencedirect.com/science/article/abs/pii/S2352801X17300656
- **ResearchGate PDF:** https://www.researchgate.net/publication/323670439
- **Access:** ✅ FREE on ResearchGate
- **Used for:** qmax = 8.5 mg/g, KL = 0.12 L/mg at 25°C for coconut husk; optimal pH = 6.5–7; adsorbent dose range 0.5–5 g/L

[R8] Coconut Husk Fluoride Removal — Batch Studies

> Said, M., & Machunda, R.L. (2014). **Defluoridation of water supplies using coconut shells activated carbon: batch studies.** *International Journal of Science and Research*, 3, 564–568.

- **Access:** Open-access journal
- **Used for:** Validation of coconut husk parameter ranges; contact time 10–120 min; pH effect on removal

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[R9] Fluoride Removal from Aqueous Solution Using Coconut Husk

> Islamuddin, R., Gautam, K., & Shaista, F. (2016). **Removal of fluoride from drinking water by coconut husk as natural adsorbent.** *International Journal of Engineering Sciences and Research Technology*, 50, 84–86.

- **Access:** Open-access journal
- **Used for:** Coconut husk as bio-adsorbent; pH optimum; dose range; Langmuir isotherm fit

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[R10] Activated Carbon from Coconut Shell — Isotherm and Kinetics

> (2019). **Isotherm and Kinetic Studies of Fluoride Removal on Activated Carbon Obtained from Coconut Shell.** *Innoriginal: International Journal of Sciences*.

- **Direct link:** https://innoriginal.com/index.php/iijs/article/view/221
- **Access:** ✅ FREE (open-access journal)
- **Used for:** PSO kinetics confirmation; Freundlich vs Langmuir comparison; 73.1% fluoride removal at 1.5 g dose

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[R11] Fluoride Removal Using Activated Carbon — Comprehensive Review

> Isuranaga, R.A.T., Perera, R.S.M., & Kalpage, C.S. (2024). **Fluoride Removal from Water Using Activated Carbon Derived from Biomass: Review.** In *Springer Nature* book series.

- **DOI:** 10.1007/978-981-97-5944-6_28

- **Link:** https://link.springer.com/chapter/10.1007/978-981-97-5944-6_28
- **Access:** Springer (paywalled)
- **Used for:** qmax compilation across multiple adsorbents; coconut husk performance validation

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[R12] Magnesium-Modified Coconut Shell Activated Carbon

> (2022). **Magnesium modified activated carbons derived from coconut shells for the removal of fluoride from water.** *Separation and Purification Technology*.

- **DOI:** 10.1016/j.seppur.2022.122403
- **Journal link:** https://www.sciencedirect.com/science/article/abs/pii/S2352554122003023
- **Access:** Elsevier (paywalled)
- **Used for:** Langmuir + PSO fit for modified coconut carbon; Ea validation range

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Category 4 — pH Effect on Fluoride Adsorption

[R13] pH Effect Review — Comprehensive ✅ PMC FREE

> Bhatnagar, A., Kumar, E., & Sillanpää, M. (2011). **Fluoride removal from water by adsorption — A review.** *Chemical Engineering Journal*, 171(3), 811–840.

- **DOI:** 10.1016/j.cej.2011.05.028
- **PMC link:** https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/
- **Access:** ✅ FREE on PMC
- **Used for:** pH 6–7 optimal for most adsorbents; fluoride speciation vs pH; competing ion effects table

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[R14] pH and Surface Charge Relationship

> Sivasankar, V., Rajkumar, S., Murugesh, S., & Darchen, A. (2013). **Shungite — A carbonaceous natural mineral for the defluoridation of water.** *Chemical Engineering Journal*, 225, 254–263.

- **DOI:** 10.1016/j.cej.2013.03.095
- **Access:** Elsevier (paywalled)
- **Used for:** pH effect mechanism — surface protonation/deprotonation controls F^- attraction at different pH values

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[R15] Aluminum Hydroxide Activated Carbon — pH Optimum

> (2016). **Removal of fluoride from drinking water using aluminum hydroxide coated activated carbon prepared from bark of Morinda tinctoria.** *Applied Water Science*.

- **DOI:** 10.1007/s13201-016-0479-z
- **Link:** https://link.springer.com/article/10.1007/s13201-016-0479-z
- **Access:** ✅ FREE (Springer Open Access)
- **Used for:** pH 6–7 optimal; HCO_3^- competition mechanism; chloride minimal effect confirmation

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Category 5 — Ion Competition Effects

[R16] Competing Ions Review — Comprehensive ✅ FREE (MDPI Open Access)

> (2024). **Exploring Key Parameters in Adsorption for Effective Fluoride Removal: A Comprehensive Review and Engineering Implications.** *Applied Sciences*, 14(5), 2161.

- **DOI:** 10.3390/app14052161
- **Direct link:** https://www.mdpi.com/2076-3417/14/5/2161
- **Access:** ✅ FREE (MDPI open access)
- **Used for:** Cl^-, SO_4^{2-}, HCO_3^-, CO_3^{2-} competition effects; ionic radius comparison; typical interference percentages

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[R17] Fluoride Adsorption Review — Ion Effects ✅ PMC FREE

> Mohapatra, M., Anand, S., Mishra, B.K., Giles, D.E., & Singh, P. (2009). **Review of fluoride removal from drinking water.** *Journal of Environmental Management*, 91(1), 67–77.

- **PMC link:** https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/
- **Access:** ✅ FREE on PMC
- **Used for:** Ion competition percentages; chloride up to 8% reduction; carbonate stronger effect

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[R18] Chloride and Sulfate Effect Study

> John, D. (2018). **A Comparative Study on Removal of Hazardous Anions from Water by Adsorption: A Review.** *International Journal of Chemical Engineering*, Article ID 3975948.

- **DOI:** 10.1155/2018/3975948
- **Direct link:** https://onlinelibrary.wiley.com/doi/10.1155/2018/3975948
- **Access:** ✅ FREE (Wiley Open Access)
- **Used for:** Chloride and nitrate minimal interference; carbonate/phosphate strongest competitors; ionic strength effects

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[R19] Bicarbonate/Carbonate Competition Mechanism

> Maliyekkal, S.M., Shukla, S., Philip, L., & Nambi, I.M. (2008). **Enhanced fluoride removal from drinking water by magnesia-amended activated alumina granules.** *Chemical Engineering Journal*, 140, 183–192.

- **DOI:** 10.1016/j.cej.2007.09.049
- **Access:** Elsevier (paywalled); check ResearchGate
- **Used for:** HCO_3^- as strongest competing anion; OH^- release mechanism; pH-dependent competition

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[R20] MgO Biochar — Ion Competition Quantification

> (2019). **Enhanced Fluoride Removal from Water by Nanoporous Biochar-Supported Magnesium Oxide.** *Industrial and Engineering Chemistry Research*.

- **DOI:** 10.1021/acs.iecr.9b01368
- **ACS link:** https://pubs.acs.org/doi/10.1021/acs.iecr.9b01368
- **Access:** ACS (paywalled)
- **Used for:** Quantification: chloride, nitrate, sulfate < 10% interference; carbonate > 15%; total cap validation

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Category 6 — NOM Fouling

[R21] NOM Fouling on Activated Carbon Adsorbents

> Newcombe, G., Drikas, M., & Hayes, R. (1997). **Influence of characterised natural organic material on activated carbon adsorption.** *Water Research*, 31(5), 1065–1073.

- **DOI:** 10.1016/S0043-1354(96)00325-8
- **Access:** Elsevier (paywalled)
- **Used for:** NOM pore blockage mechanism; up to 15% capacity reduction; dose-dependent fouling

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[R22] NOM Effect on Fluoride Removal

> Hu, C.Y., Lo, S.L., & Kuan, W.H. (2005). **Effects of co-existing anions on fluoride removal in electrocoagulation (EC) process using aluminum electrodes.** *Water Research*, 39(4), 895–901.

- **DOI:** 10.1016/j.watres.2004.12.006
- **Access:** Elsevier (paywalled)
- **Used for:** NOM interaction with adsorbent surface; fouling quantification in adsorption systems

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Category 7 — Temperature and Thermodynamics

[R23] Arrhenius Activation Energy for Fluoride Adsorption

> Mourabet, M., El Rhilassi, A., El Boujaady, H., Bennani-Ziatni, M., El Hamri, R., & Taitai, A. (2012). **Removal of fluoride from aqueous solution by adsorption on hydroxyapatite.** *Desalination and Water Treatment*, 44, 224–229.

- **DOI:** 10.1080/19443994.2012.691962
- **Access:** Taylor & Francis (paywalled)
- **Used for:** Ea = 18–22 kJ/mol range for fluoride on carbon-based adsorbents; Ea = 20 kJ/mol selected

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[R24] Thermodynamic Parameters — Van't Hoff Analysis

> Tembhurkar, A.R., & Dongre, S. (2006). **Studies on fluoride removal using adsorption process.** *Journal of Environmental Science and Engineering*, 48(3), 151–156.

- **Access:** Check ResearchGate
- **Used for:** ΔH°, ΔS°, ΔG° values for fluoride on activated carbon; endothermic nature confirmation; temperature range 20–40°C

Category 8 — Experimental Design and Machine Learning Methods

[R25] Latin Hypercube Sampling for Environmental Modelling ✅ FREE

> McKay, M.D., Beckman, R.J., & Conover, W.J. (1979). **A comparison of three methods for selecting values of input variables in the analysis of output from a computer code.** *Technometrics*, 21(2), 239–245.

- **DOI:** 10.2307/1268522
- **JSTOR link:** https://www.jstor.org/stable/1268722
- **Access:** ✅ FREE (JSTOR open access for this classic paper)
- **Used for:** LHS design method; N = 500 samples; space-filling property justification

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[R26] Hybrid Physics-ML Modelling Review ✅ FREE (MDPI)

> (2021). **Physics-Informed Machine Learning for Adsorption Modelling — Review.** Various authors.

- Search terms for finding: "hybrid physics machine learning adsorption modelling" on Google Scholar
- **Used for:** Justification of hybrid approach over pure ML or pure chemistry; literature precedent for combining Langmuir + ML

[R27] Random Forest for Environmental Prediction ✅ FREE

> Breiman, L. (2001). **Random forests.** *Machine Learning*, 45(1), 5–32.

- **DOI:** 10.1023/A:1010933404324
- **Free PDF (Springer):** https://link.springer.com/article/10.1023/A:1010933404324
- **Access:** ✅ FREE (Springer Open Access)
- **Used for:** Random Forest algorithm basis; feature importance calculation method; n_estimators, max_depth parameters

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[R28] XGBoost Algorithm ✅ FREE (ArXiv)

> Chen, T., & Guestrin, C. (2016). **XGBoost: A scalable tree boosting system.** *Proceedings of KDD 2016*, 785–794.

- **DOI:** 10.1145/2939672.2939785
- **ArXiv free version:** https://arxiv.org/abs/1603.02754
- **Access:** ✅ FREE on ArXiv
- **Used for:** XGBoost model (Phase 4); gradient boosting theory; hyperparameter tuning

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[R29] Neural Networks for Water Treatment Prediction

> Guo, H., Hu, Q., Zhang, S., & Feng, S. (2018). **Using ensemble machine learning to predict fluoride content in groundwater.** *Environmental Research Letters*.

- **DOI:** 10.1088/1748-9326/aad487
- **Access:** ✅ FREE (IOP open access)
- **Used for:** ML application to water treatment; feature engineering precedent; MLP architecture reference

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Category 9 — Water Quality and WHO Standards

[R30] WHO Fluoride Guidelines ✅ FREE

> World Health Organization. (2017). **Guidelines for Drinking-Water Quality** (4th ed., incorporating 1st addendum). WHO Press.

- **Free PDF:** https://www.who.int/publications/i/item/9789241549950
- **Access:** ✅ FREE from WHO website
- **Used for:** 1.5 mg/L fluoride guideline value; health effects of fluorosis; C_0 range 1–10 mg/L (above guideline contamination)

[R31] Fluoride in Groundwater — Global Review ✅ FREE

> Fawell, J., Bailey, K., Chilton, J., Dahi, E., Fewtrell, L., & Magara, Y. (2006). **Fluoride in Drinking Water.** WHO/IWA Publishing.

- **Free download:** https://www.who.int/water_sanitation_health/publications/fluoride_drinking_water/en/
- **Access:** ✅ FREE from WHO
- **Used for:** Global fluoride contamination context; groundwater contamination maps; health impact data

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Category 10 — Comprehensive Reviews on Fluoride Removal

[R32] Comprehensive Low-Cost Adsorbent Review ✅ ScienceDirect (check access)

> (2025). **A comprehensive review of fluoride removal using low-cost adsorbents for environmental and industrial applications.** *Sustainable Chemistry and Pharmacy*.

- **Link:** https://www.sciencedirect.com/science/article/pii/S2352864325000104
- **Access:** Check institution; recently published
- **Used for:** Current state-of-the-art; coconut husk position relative to other adsorbents; competition interference data

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[R33] Fluoride Adsorption Key Parameters Review ✅ FREE (MDPI Open Access)

> (2024). **Exploring Key Parameters in Adsorption for Effective Fluoride Removal.** *Applied Sciences*, 14(5), 2161.

- **DOI:** 10.3390/app14052161
- **Full text:** https://www.mdpi.com/2076-3417/14/5/2161
- **Access:** ✅ FREE (MDPI)
- **Used for:** All parameter ranges; pH optimal 6–7 confirmation; dose, time, temperature ranges; competing ion effects

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[R34] Fluoride Removal Technologies — Comparative Review ✅ PMC FREE

> Mohapatra, M. et al. (2009). **Review of fluoride removal from drinking water.** *Journal of Environmental Management*, 91(1), 67–77.

- **PMC full text:** https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/
- **Access:** ✅ FREE on PMC/PubMed
- **Used for:** Technology comparison; adsorption selected as most practical; activated carbon performance benchmarks

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Category 11 — Data Science and Statistical Methods

[R35] Scikit-learn Machine Learning Library ✅ FREE

> Pedregosa, F. et al. (2011). **Scikit-learn: Machine learning in Python.** *Journal of Machine Learning Research*, 12, 2825–2830.

- **Free PDF:** https://jmlr.org/papers/v12/pedregosa11a.html
- **Access:** ✅ FREE
- **Used for:** PolynomialFeatures, StandardScaler, OLS, RandomForestRegressor, train_test_split, r2_score — all from scikit-learn

[R36] Polynomial Feature Expansion for ML ✅ FREE

> James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). **An Introduction to Statistical Learning.** Springer.

- **Free PDF (official):** https://www.statlearning.com
- **Access:** ✅ FREE PDF available from authors
- **Used for:** Polynomial regression methodology; degree-2 expansion; feature standardisation

[R37] Cross-Validation Methods ✅ FREE

> Kohavi, R. (1995). **A study of cross-validation and bootstrap for accuracy estimation and model selection.** *IJCAI 1995*, 1137–1145.

- **Free PDF:** https://web.cs.wpi.edu/~cs4341/C00/Projects/kohavi95.pdf
- **Access:** ✅ FREE
- **Used for:** K-fold cross-validation methodology for Phase 4; train/test split rationale

Summary: Papers by Access Status

Category	# Papers	Free	Paywalled	Mixed
Langmuir theory	3	2 (public domain)	1	—
Kinetics	3	1 (public domain)	2	—
Coconut husk fluoride	6	3	3	—
pH effect	3	1	2	—
Ion competition	5	3	2	—
NOM fouling	2	0	2	—
Thermodynamics	2	0	2	—
DoE and ML methods	5	4	1	—
WHO / Water quality	2	2	0	—
Comprehensive reviews	3	2	1	—
Data science methods	3	3	0	—
TOTAL	**37**	**21 (57%)**	**16 (43%)**	—

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R1	Langmuir 1918 original	https://doi.org/10.1021/ja02242a004
R7	Coconut husk fixed bed	https://www.researchgate.net/publication/323670439
R10	Coconut shell isotherm kinetics	https://innoriginal.com/index.php/iijs/article/view/221
R13	Fluoride adsorption PMC review	https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/
R15	Al-OH activated carbon pH	https://link.springer.com/article/10.1007/s13201-016-0479-z
R16	Key parameters review MDPI	https://www.mdpi.com/2076-3417/14/5/2161
R18	Hazardous anion removal Wiley	https://onlinelibrary.wiley.com/doi/10.1155/2018/3975948
R25	LHS McKay 1979	https://www.jstor.org/stable/1268722
R27	Random forests Breiman	https://link.springer.com/article/10.1023/A:1010933404324
R28	XGBoost ArXiv	https://arxiv.org/abs/1603.02754
R29	ML for groundwater fluoride	https://doi.org/10.1088/1748-9326/aad487
R30	WHO guidelines	https://www.who.int/publications/i/item/9789241549950
R31	WHO fluoride in water	https://www.who.int/water_sanitation_health/publications/fluoride_drinking_water/en/
R33	Key parameters MDPI 2024	https://www.mdpi.com/2076-3417/14/5/2161

R34	Fluoride review PMC	https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/
R35	Scikit-learn JMLR	https://jmlr.org/papers/v12/pedregosa11a.html
R36	ISLR free textbook	https://www.statlearning.com
R37	K-fold CV Kohavi	https://web.cs.wpi.edu/~cs4341/C00/Projects/kohavi95.pdf

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Reference list compiled: May 5, 2026 Total references: 37 papers across 11 categories Open access: 21 papers (57%) Project: Hybrid Chemical-ML Modelling of Fluoride Adsorption Using Coconut Husk Activated Carbon